

Erwin Quanten

Staff Software Engineer – Python, Django, FastAPI, AWS

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PROFESSIONAL SUMMARY

Python Backend Engineer with strong experience building API-driven platforms for logistics, digital consulting, chemicals, manufacturing, and enterprise operations using **Python 3.8–3.12**, **FastAPI 0.95–0.111**, **Django 3.2–5.0**, **PostgreSQL 12–16**, and **AWS**. Skilled in designing scalable REST services, async workers, data pipelines, authentication flows, containerized deployments, and production monitoring. Known for migrating legacy services, improving response time by 18–26%, reducing failed jobs by 21–28%, and delivering reliable backend features with clean architecture, practical debugging, secure coding, and flexible collaboration across product, DevOps, and QA teams.

SKILLS

- **Backend: Python 3.8–3.12, FastAPI 0.95–0.111, Django 3.2–5.0**, Django REST Framework 3.12–3.15, Flask 1.1–2.3, Pydantic 1.10–2.7, SQLAlchemy 1.4–2.0
- **Database: PostgreSQL 12–16**, MySQL 8, Redis 6–7, MongoDB 4.4–6.0, Alembic, Django ORM, query optimization, indexing, migrations
- **Cloud & DevOps: AWS**, Docker 20–25, Kubernetes 1.22–1.29, GitHub Actions, GitLab CI, Terraform 1.2–1.7, Nginx, Linux
- **Messaging & Async: Celery 5**, RabbitMQ 3.8–3.12, AWS SQS, Redis Queue, background jobs, retries, dead-letter queues
- **Testing & Quality: Pytest 6–8**, unittest, Factory Boy, Postman, OpenAPI, Swagger, contract testing, integration testing
- **Architecture: REST APIs**, microservices, modular monoliths, event-driven systems, OAuth2, JWT, RBAC, clean architecture, observability

EXPERIENCE

Lightedge — Lead Software Engineer

Remote/ Full time

May2022 – Apr 2026

Houston, TX

- Led backend architecture for logistics and infrastructure platforms using **Python 3.10–3.12**, **FastAPI 0.95–0.111**, **PostgreSQL 14–16**, and **AWS ECS**.
- Designed high-throughput REST APIs with **FastAPI**, Pydantic validation, dependency injection, and OpenAPI contracts to support customer portals and internal operations.
- Improved API response time by **24%** after profiling slow endpoints, rewriting inefficient ORM access, and adding targeted PostgreSQL indexes.
- Reduced background job failures by **27%** by introducing Celery retry policies, dead-letter queues, idempotency keys, and clearer operational alerts.
- Migrated legacy Flask services into a modern **FastAPI** structure with service layers, typed schemas, async database access, and reusable authentication middleware.
- Implemented secure OAuth2 and JWT authentication with role-based permissions, token refresh logic, audit tracking, and tenant-aware authorization rules.
- Resolved production timeout issues caused by unbounded database joins by splitting queries, adding pagination, and caching heavy read models in Redis.
- Built asynchronous workers for invoice processing, customer notifications, log aggregation, and infrastructure events using **Celery 5**, RabbitMQ, and Redis.
- Improved deployment stability by **22%** through Docker image optimization, health checks, rollback scripts, and GitHub Actions validation before production release.
- Created reusable backend modules for file uploads, audit logs, permissions, email delivery, and API error handling to speed up feature development.
- Partnered with frontend, DevOps, QA, and product teams to deliver backend features flexibly while keeping code reviews strict and release risk controlled.
- Standardized Pytest integration tests, fixture factories, and API contract tests so backend releases could move faster without increasing regression defects.

Syscom Digital — Staff Software Engineer

Remote/ Full time

Nov 2017 – Apr 2022

Bucharest, Romania

- Built enterprise backend systems for digital consulting clients using **Python 3.7–3.10, Django 2.2–4.0, FastAPI 0.63–0.85**, and PostgreSQL.
- Developed REST APIs for customer onboarding, document workflows, reporting dashboards, and third-party integrations with clean service boundaries.
- Reduced report generation time by **23%** by replacing synchronous queries with Celery jobs, optimized materialized views, and Redis-based progress tracking.
- Improved API reliability by **19%** after adding structured exception handling, request tracing, correlation IDs, and consistent JSON error responses.
- Migrated older Django modules from function-based views to class-based APIs with Django REST Framework serializers, permissions, pagination, and throttling.
- Implemented integration features with payment providers, CRM systems, email services, and internal ERP platforms using resilient retry and timeout patterns.
- Fixed recurring database lock issues by analyzing transaction scope, reducing long-running updates, and separating read-heavy analytics from write paths.
- Created backend templates for new services, including Docker setup, environment validation, logging configuration, testing structure, and CI pipeline checks.
- Guided engineers on **Python**, Django ORM optimization, API versioning, and secure coding practices while keeping delivery aligned with client deadlines.
- Supported production incidents by reading logs, reproducing edge cases, preparing hotfixes, and documenting root causes for future prevention.

Tessenderlo Group — Senior Software Engineer

Remote/ Full time

Aug 2015 – Oct 2017

Brussel Metropolitan Area

- Delivered backend services for chemicals, manufacturing, and supply-chain operations using **Python 3.5–3.7, Django 1.11–2.0**, Flask, and PostgreSQL.
- Built APIs for production planning, inventory visibility, shipment tracking, and internal reporting with practical validation and database transaction control.
- Improved batch import performance by **21%** by replacing row-by-row processing with bulk inserts, staged validation, and better PostgreSQL indexing.
- Reduced operational data errors by **18%** after adding stricter server-side validation, reconciliation scripts, and exception reports for business users.
- Migrated legacy Python scripts into maintainable Django management commands with reusable service functions, logging, scheduling, and rollback support.
- Implemented new backend features for approval workflows, plant-level reporting, supplier updates, and secure internal user access management.
- Solved recurring CSV import failures caused by encoding and delimiter inconsistencies by adding auto-detection, validation summaries, and safer error handling.
- Worked with infrastructure teams to containerize backend services, separate environment configuration, and prepare cleaner deployment packages.
- Mentored team members on debugging production issues, writing safer migrations, and using backend skills flexibly across business-critical systems.

Tessenderlo Group — Software Engineer

Hybrid/ Full time

Sep 2011 – Jul 2015

Brussel Metropolitan Area

- Developed internal business applications using **Python 2.7–3.5, Django 1.4–1.8**, PostgreSQL, jQuery, and Linux-based deployment environments.
- Created backend modules for procurement requests, stock movements, production forms, and operational reporting used by multiple internal departments.
- Improved monthly reporting speed by **20%** by moving heavy calculations from application loops into optimized SQL queries and scheduled jobs.
- Reduced support tickets by **16%** after adding clearer validation messages, better form handling, and automatic checks for duplicate business records.
- Migrated several legacy scripts from manual execution into scheduled backend jobs with logging, email notifications, and basic failure recovery.

- Implemented new features for role-based access, document attachments, approval status tracking, and searchable operational history across departments.
- Fixed bugs related to timezone handling, broken file paths, missing database constraints, and inconsistent user permissions in production workflows.
- Collaborated with analysts and operations users to translate business rules into stable backend logic without overcomplicating the application design.

Tessengerlo Group — Software Developer

Hybrid/ Full time

Jun 2009 – Aug 2011

Brussel Metropolitan Area

- Supported internal software tools using **Python 2.6–2.7**, early Django versions, SQL, HTML, CSS, JavaScript, and Linux server environments.
- Maintained backend forms, reporting pages, database queries, and small automation scripts for manufacturing, finance, and administrative teams.
- Improved small automation scripts by **14%** in runtime after simplifying loops, removing duplicate queries, and adding safer input validation.
- Fixed production issues involving broken reports, incorrect calculations, missing field validation, and permission mismatches reported by business users.
- Created documentation for deployment steps, recurring fixes, database scripts, and support workflows so senior engineers could review and improve them faster.

EDUCATION

KU Leuven — Master's Degree, Computer Science

Sep 2003 – May 2009